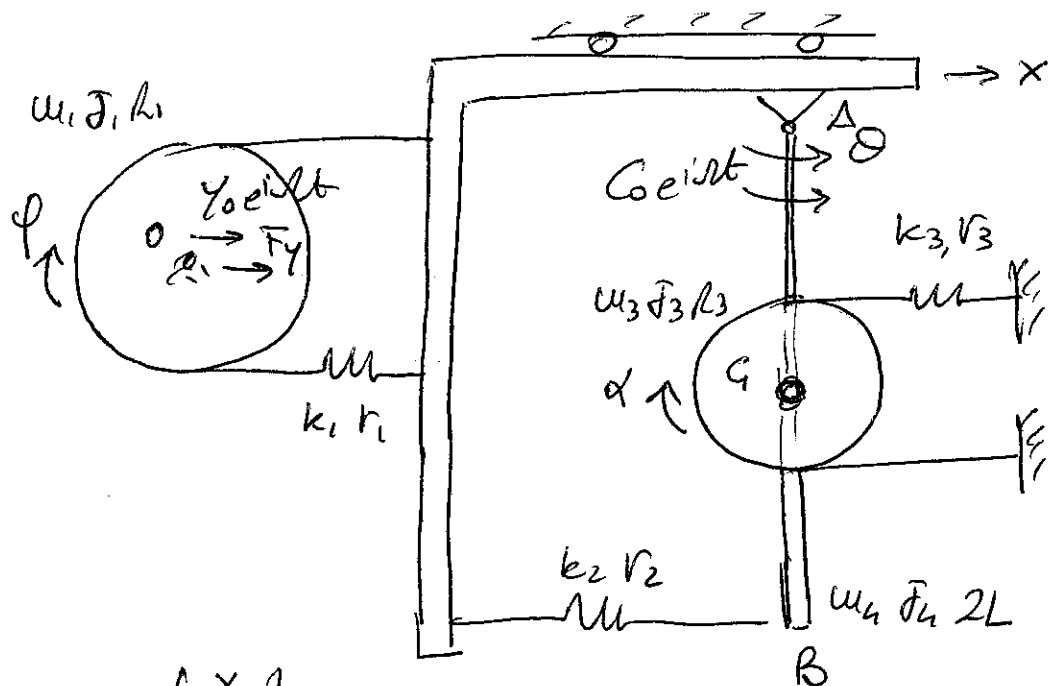


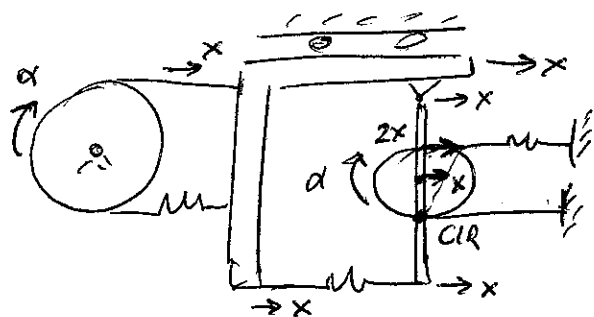


$$x = \begin{Bmatrix} x \\ \theta \\ y \end{Bmatrix}$$

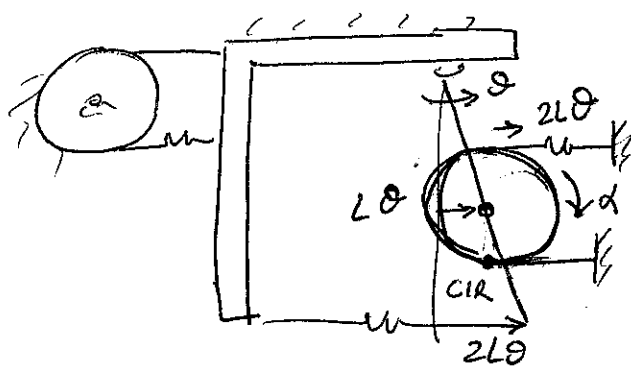
$$E_c = \frac{1}{2} m_1 \dot{y}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}^2 + \frac{1}{2} (m_3 + m_4) \dot{x}_s^2 + \frac{1}{2} J_3 \dot{\alpha}^2 + \frac{1}{2} J_4 \dot{\theta}^2$$

$$V_{el} = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2 + \frac{1}{2} k_3 \Delta L_3^2$$

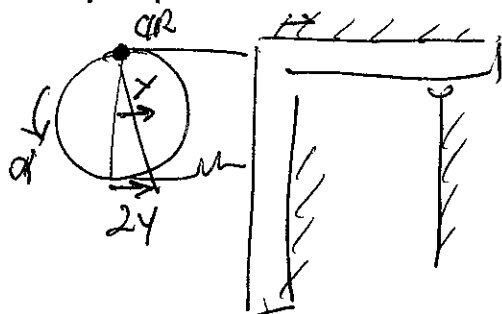
$$x \neq 0; \theta = y = 0$$



$$x = y = 0; \theta \neq 0$$



$$y \neq 0; x = \theta = 0$$



$$\varphi = \frac{x}{r_1} - \frac{y}{r_1}$$

$$x_s = x + L\theta$$

$$\alpha = \frac{x}{r_3} + \frac{L\theta}{r_3}$$

$$\Delta L_1 = 2x - 2y$$

$$\Delta L_2 = 2L\theta$$

$$\Delta L_3 = -2x - 2L\theta$$

$$u_F = \text{diag}(u_1, \bar{F}_1, u_2, u_3+u_4, \bar{F}_3, \bar{F}_4)$$

$$k_F = \text{diag}(k_1, k_2, k_3)$$

$$r_F = \text{diag}(r_1, r_2, r_3)$$

$$\Delta u = \begin{bmatrix} 0 & 0 & 1 \\ 1/r_1 & 0 & -1/r_1 \\ 1 & 0 & 0 \\ 1 & L & 0 \\ 1/r_3 & 4/r_3 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

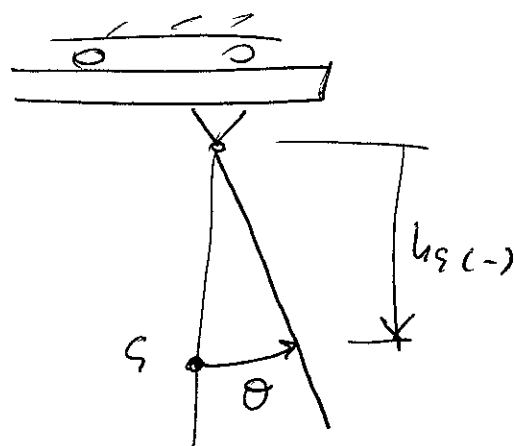
$$\Delta k = \Delta r = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 2L & 0 \\ -2 & -2L & 0 \end{bmatrix}$$

$$V_s = (u_3 + u_4) g h_s$$

$$h_s = -L \cos \theta$$

$$\frac{\partial^2 V_s}{\partial \theta^2} = (u_3 + u_4) g L \cos \theta$$

$$k_s = \begin{bmatrix} 0 & 0 & 0 \\ 0 & (u_3 + u_4) g L & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

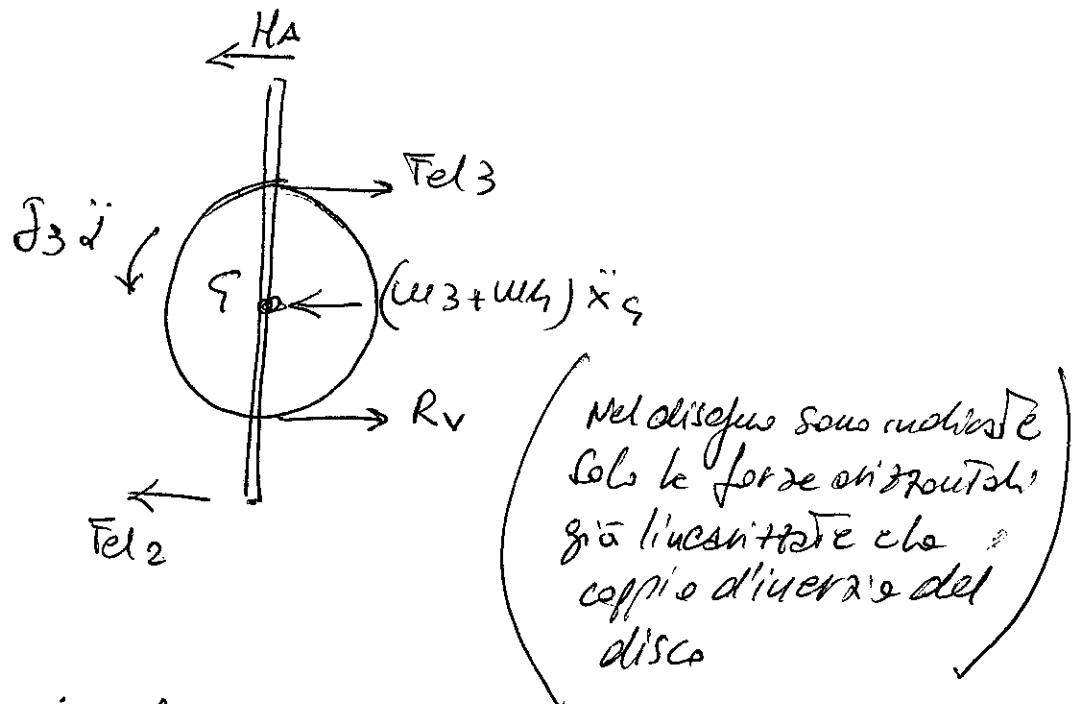


$$\delta L = C_0 e^{int} \delta \theta + F_y \delta y$$

$$\rightarrow Q_c = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} C_0 e^{int} \quad Q_y = \begin{Bmatrix} 0 \\ 0 \\ F_y \end{Bmatrix}$$

$$F_{el3} = (k_3 + i\omega \nu_3) \Delta L_3 \quad (\text{domanda } L_3)$$

$$F_y = Q_v = [M_{vL}] \ddot{x}_L + [R_{vL}] \dot{x}_L + [C_{vL}] x_L + [M_{vv}] \ddot{x}_v + [C_{vv}] \dot{x}_v + [K_{vv}] x_v$$



$$\sum M_G^{DISCO} = 0 : \quad R_v R_3 + J_3 \ddot{\theta} - F_{el3} R_3 = 0 \rightarrow$$

$$\rightarrow \boxed{R_v = F_{el3} - \frac{J_3 \ddot{\theta}}{R_3}}$$

$$\sum F_x = 0 : \quad F_{el3} + R_v - H_A - (w_3 + w_4) \ddot{x}_s - F_{el2} = 0$$

$$\boxed{H_A = F_{el3} + R_v - (w_3 + w_4) \ddot{x}_s - F_{el2}}$$

$$F_{el2} = (k_2 + i\omega \nu_2) \Delta L_2$$

$$F_{el3} = (k_3 + i\omega \nu_3) \Delta L_3$$

$$\ddot{x}_s = -\omega^2 x_s$$

positive per trazione delle molle $\rightarrow$
 $\rightarrow$ sullo struttura applicate con il
 principio di azione e reazione